Functions and relations

1 Express the relation $\{(2,2),(4,4),(6,3),(7,5)\}$ in a table of values.

2 Express the relation $\{(1,2),(3,5),(7,8),(8,11)\}$ as a series of points on the coordinate plane.

3 Determine whether the following statements are true or false.

a When working with a function, substituting a certain value of x into the formula gives only one value of y for that value of x.

b All relations are functions.

c Some relations are functions.

d All functions are relations.

e No functions are relations.

f No relations are functions.

g A relation always passes the vertical line test.

h The graph of every straight line is a function.

i The graph of every non-vertical straight line is a function.

j There is no straight line graph that is a function.

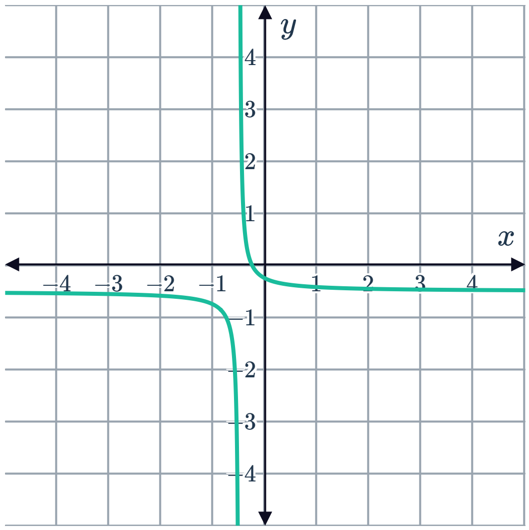
k The graph of every straight line is a relation.

l The graph of every non-horizontal straight line is a function.

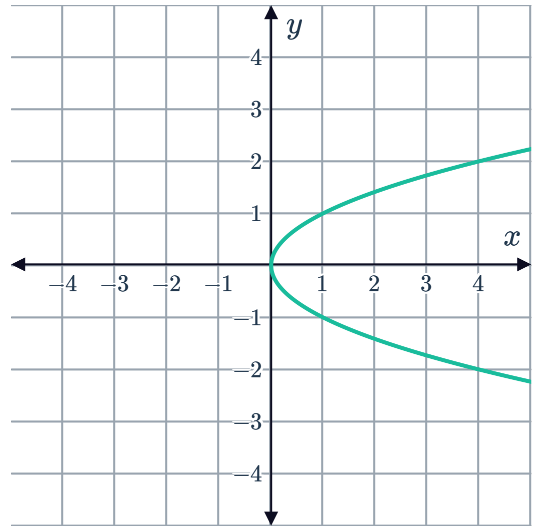
4 Identify the name used to describe a graph where for some value of x, there exists two or more different values of y.

5 Determine whether each of the following g[redacted] describes a relation only or both a relation and a function.

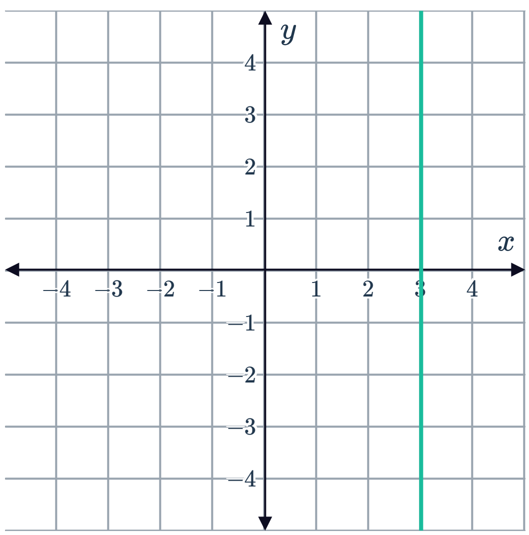
a



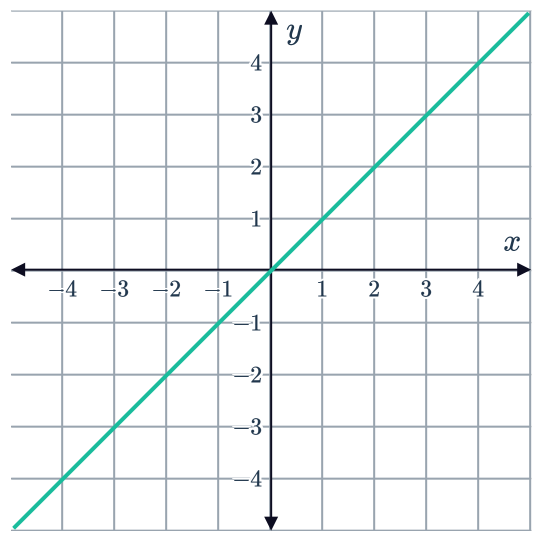
b



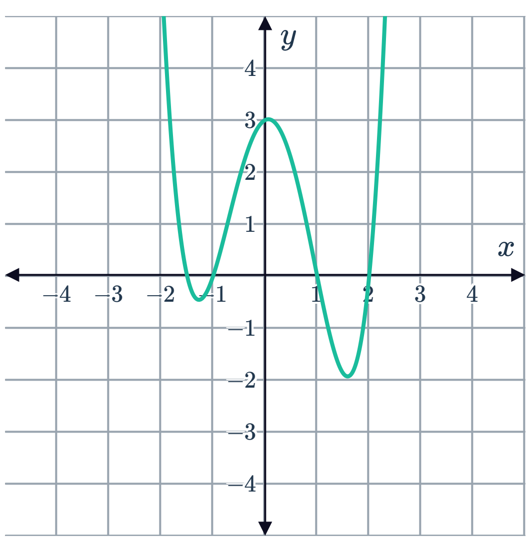
c



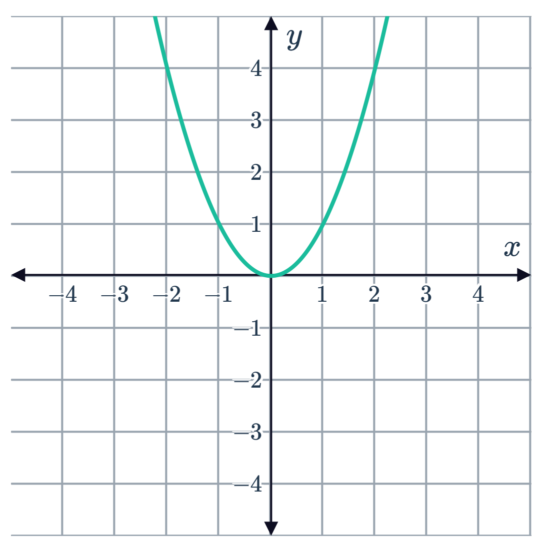
d



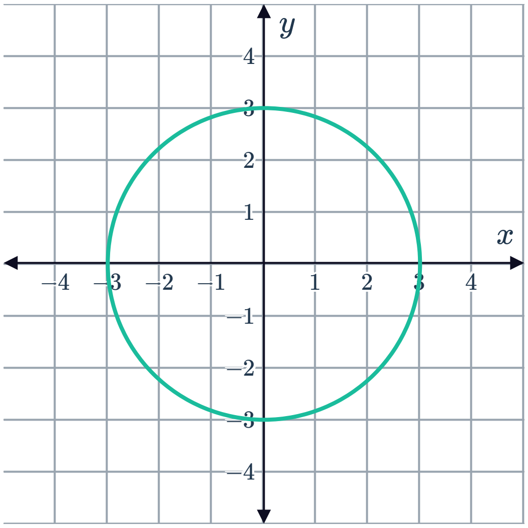
e

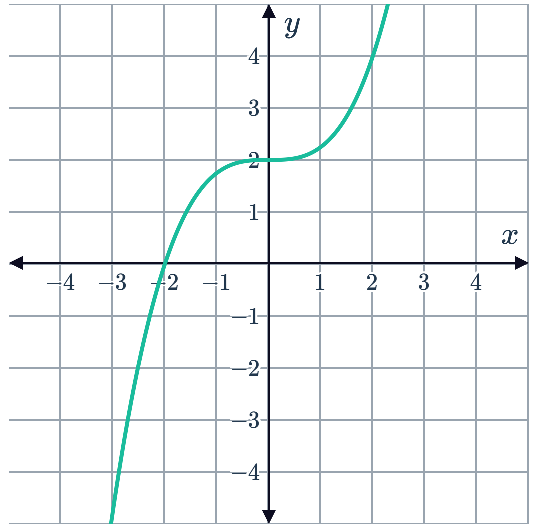


f

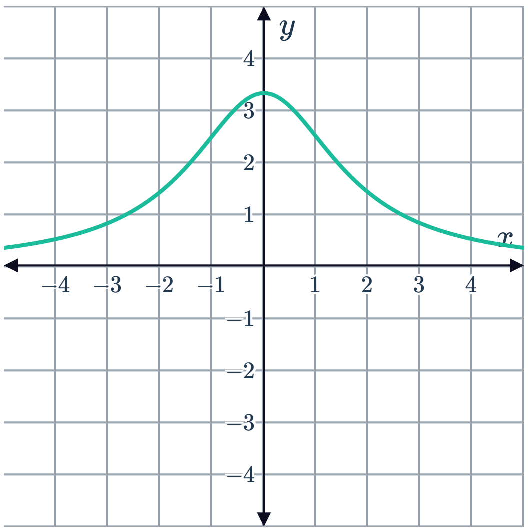


g

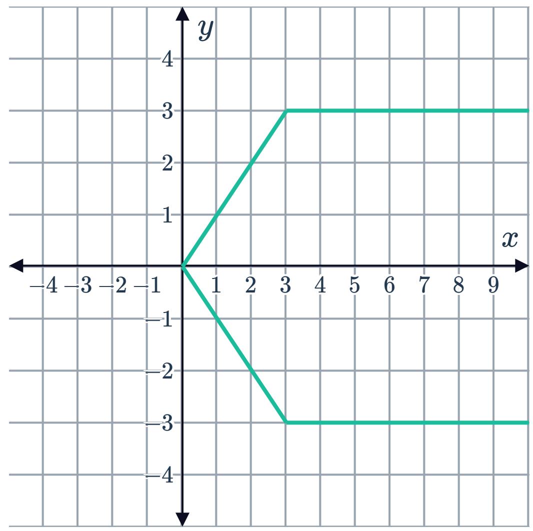




i



j



6 Consider the following set of points:
$\{(16,14),(8,28),(-15,14),(-20,-28)\}$

- **a** List the domain of the set of points.
- **b** List the range of the set of points.
- **c** Sta whether the points describe a function.

7 State whether the following set of points describe a function, a relation, both, or neither.

- **a** $\{(2,5),(7,-3),(5,2),(-4,-9)\}$
- **b** $\{(2,5),(2,7),(-3,-4),(-9,13)\}$
- **c** $(1,5),(1,1),(7,-2),(-5,-10)$
- **d** $(1,5),(7,-2),(-5,-10),(13,-13)$
- **e** $(1,5),(1,7),(-2,-5),(-5,-10)$

8 The pairs of values in the table represent a relation between x and y. Determine whether they represent a function.

a

x	-4	-3	-2	-1	0	1	2	3	4
y	-4	-3	-2	-1	0	-1	-2	-3	-4

b

x	0	1	4	8	9	12	16	18	20
y	0	1	2	$2\sqrt{2}$	3	$2\sqrt{3}$	4	$3\sqrt{2}$	$2\sqrt{5}$

c

x	-9	-7	-6	-5	-3	-2	3	5	10
y	10	10	10	10	10	10	10	10	10

d

x	-8	-7	-6	-3	2	7	9	9	10
y	8	13	-18	-16	-15	-2	-4	11	-9

9 Consider the points in the table.

x	-4	-3	0	-1	0	1	2	3	4
y	4	3	2	1	0	1	2	3	4

a Plot the points on the coordinate plane.

b State whether the relation is a function.

10 A relation is defined as follows:
$y = 1$ if x is positive and $y = -1$ if x is 0 or negative.

a Complete the following table.

x	-4	-3	-2	-1	0	1	2	3	4
y									

b Plot the points on the coordinate plane.

c State whether the relation is a function.

11 Dodgi Mobile charges $\$2.90$ a minute plus a connection fee of $\$0.60$ for an international call.

a Complete the table.

Call length (minutes)	International Call cost
1	
2	
3	
4	
5	

b State whether the relation is a function.

12 A particular store is offering one free T-shirt for every two T-shirts purchased, where each T-shirt costs $\$19$.

a Complete the table.

Number of T-shirts	Total Cost (dollars)
1	
2	
3	
4	
5	

b State whether the relation is a function.

13 Consider these ordered pairs:
$\{(-9,-5),(-5,-10),(-5,-4),(-3,7),(-2,-4),(-1,1)\}$

a Plot the ordered pairs on the number plane.

b State whether the relation is a function.

c Identify the ordered pair needed to be removed from the set so that the remaining ordered pairs represent a function.

14 Tyrah makes scarfs to sell at the market. It c[illegible]er $\$2$ to produce each one, and she sells them for $\$5$.

a Consider when 1, 2, 3, 4 and 5 scarfs are sold. Graph the points representing the relation between the number of scarfs she manages to sell and her total profit.

b State whether the relation is a function.

15 Find the possible values of k such that each of the following relations does not represent a function.

a $\{(2,1),(7,k),(5,8),(3,9)\}$

b $\{(6,2),(8,5),(1,7),(k,4)\}$

16 Determine whether each of the following table of values represents a function.

a

x	y
-4	5
2	4
1	2
1	1

b

x	y
3	-4
2	-3
1	-1
2	0

c

x	y
2	1
2	3
2	5
2	7

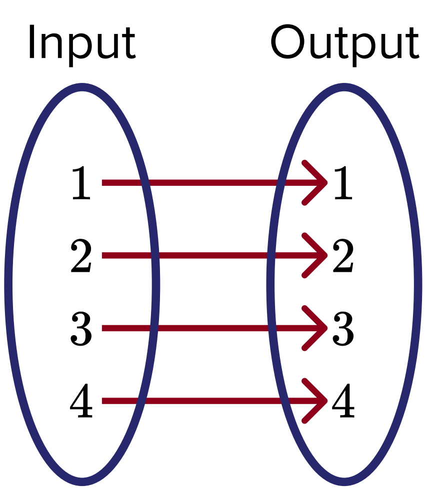
d

x	y
-1	2
0	1
1	2
2	3

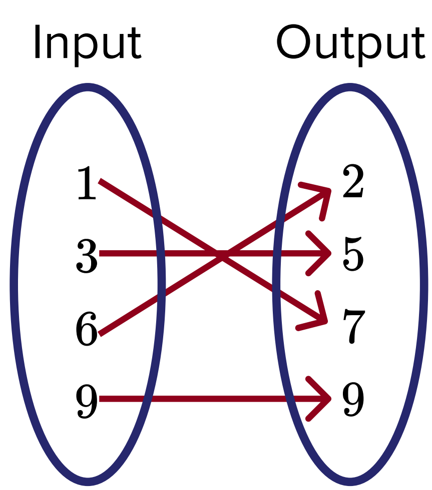
Additional questions

17 State whether each of the following mapping diagrams represents a function.

a



b



c